SQL Data Analytics project

Blinkit *Data* Analysis

A deep-dive SQL analysis of Blinkit’s operational data - uncovering trends in order patterns, delivery windows, category performance and customer behavior using structured query.

Tool :-

PostgreSQL / MYSQL workbench

Domain :-

Quick-commerce

Type :-

Analysis using SQL

Project Overview

- This project analyzes Blinkit grocery sales data to understand business performance, customer behavior, and product trends.
- The objective is to clean the dataset, ensure consistency, and prepare it for accurate analysis using SQL.

Dataset Overview

The dataset contains information related to:

- Item Fat Content
- Item Identifier
- Item Type
- Outlet Establishment Year
- Outlet Identifier
- Outlet Location Type
- Outlet Size
- Outlet Type
- Item Visibility
- Item Weight
- Total Sales
- Rating

This data is used to derive insights about sales performance and customer preferences.

1: View Raw Data

Query:

```
SELECT *
FROM blinkit_grocery_data;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size	Outlet Type	Item Visibility	Item Weight
Regular	FDX32	Fruits and Vegetables	2012	OUT049	Tier 1	Medium	Supermarket Type1	0.1000135	15.1
Low Fat	NCB42	Health and Hygiene	2022	OUT018	Tier 3	Medium	Supermarket Type2	0.008596051	11.8
Regular	FDR28	Frozen Foods	2010	OUT046	Tier 1	Small	Supermarket Type1	0.025896485	13.85
Regular	FDL50	Canned	2000	OUT013	Tier 3	High	Supermarket Type1	0.042277867	12.15
Low Fat	DRI25	Soft Drinks	2015	OUT045	Tier 2	Small	Supermarket Type1	0.033970195	19.6
Low Fat	FDS52	Frozen Foods	2020	OUT017	Tier 2	Small	Supermarket Type1	0.005505481	8.89
Low Fat	NCU05	Health and Hygiene	2011	OUT010	Tier 3	Small	Grocery Store	0.098312421	11.8
Low Fat	NCD30	Household	2015	OUT045	Tier 2	Small	Supermarket Type1	0.026903714	19.7
Low Fat	FDW20	Fruits and Vegetables	2000	OUT013	Tier 3	High	Supermarket Type1	0.024129332	20.75
Low Fat	FDX25	Canned	1998	OUT027	Tier 3	Medium	Supermarket Type3	0.101561568	NULL
Low Fat	FDX21	Snack Foods	1998	OUT027	Tier 3	Medium	Supermarket Type3	0.084554569	NULL
Low Fat	NCU41	Health and Hygiene	2017	OUT035	Tier 2	Small	Supermarket Type1	0.052044976	18.85
Low Fat	FDL20	Fruits and Vegetables	2022	OUT018	Tier 3	Medium	Supermarket Type2	0.128937661	17.1
Low Fat	NCR54	Household	2000	OUT013	Tier 3	High	Supermarket Type1	0.090486828	16.35
Low Fat	FDH19	Meat	1998	OUT027	Tier 3	Medium	Supermarket Type3	0.03292824	NULL
Regular	FDB57	Fruits and Vegetables	2017	OUT035	Tier 2	Small	Supermarket Type1	0.018801549	20.25
Low Fat	FDO23	Breads	2022	OUT018	Tier 3	Medium	Supermarket Type2	0.147023834	17.85
Low Fat	NCB07	Household	2012	OUT049	Tier 1	Medium	Supermarket Type1	0.077628053	19.2
Low Fat	FDJ56	Fruits and Vegetables	1998	OUT027	Tier 3	Medium	Supermarket Type3	0.182514881	NULL
Low Fat	DRN47	Hard Drinks	2022	OUT018	Tier 3	Medium	Supermarket Type2	0.016895293	12.1
Regular	FDZ07	Fruits and Vegetables	1998	OUT027	Tier 3	Medium	Supermarket Type3	0	NULL
Low Fat	NCK31	Others	1998	OUT027	Tier 3	Medium	Supermarket Type3	0.026916794	NULL
Low Fat	FDJ41	Frozen Foods	2022	OUT018	Tier 3	Medium	Supermarket Type2	0.022976497	6.85

Explanation:

This query is used to preview the dataset and understand its structure before performing any transformations.

2: Data Cleaning

Problem:
The Item Fat Content column contains inconsistent values like:

- LF
- low fat
- reg

This creates issues in grouping and analysis.

Query:

```sql
UPDATE blinkit_grocery_data
SET `Item Fat Content` =
CASE
    WHEN `Item Fat Content` IN ('LF', 'low fat') THEN 'Low Fat'
    WHEN `Item Fat Content` = 'reg' THEN 'Regular'
    ELSE `Item Fat Content`
END;
```

Explanation:
Standardized values to ensure consistency across the dataset.

3: Data Validation

Query:

```
SELECT DISTINCT `Item Fat Content`
FROM blinkit_grocery_data;
```

Output:

Result Grid

Item Fat Content
Regular
Low Fat

Explanation:
This step verifies that the data cleaning was successful and all values are now consistent.

A: KPI Analysis Overview

This section focuses on calculating key performance indicators (KPIs) to understand overall business performance.The KPIs analyzed include:

1. Total Sales
2. Average Sales
3. Number of Orders
4. Average Rating

5. **Total Sales**

Query:

```sql
SELECT
CAST( SUM(`Total Sales`)/1000000
AS DECIMAL(10,2))
AS Total_Sales_in_Millions
FROM blinkit_grocery_data;
```

Output:

Result Grid | Filter

	Total_Sales_in_Millions
▶	1.20

2. **Average Sales**

Query:

```
SELECT
CAST(AVG(`Total Sales`)
AS DECIMAL(10,0)) AS Avg_Sales
FROM blinkit_grocery_data;
```

Output:

Result Grid

	Avg_Sales
▶	141

3. **Number of Orders**

Query:

```
SELECT COUNT(*)
AS Total_Items
FROM blinkit_grocery_data;
```

Output:

Result Grid

	Total_Items
▶	8523

4. **Average Rating**

Query:

```sql
SELECT
CAST(AVG(`Rating`)
AS DECIMAL(10,2))
AS Avg_Rating
FROM blinkit_grocery_data;
```

Output:

Result Grid

	Avg_Rating
▶	3.96

B: Sales by Fat Content

Query:

```
SELECT `Item Fat Content`,
CAST(SUM(`Total Sales`)AS DECIMAL(10,2))AS Total_Sales
FROM blinkit_grocery_data
GROUP BY `Item Fat Content`
ORDER BY Total_Sales DESC;
```

Output:

Result Grid | Filter Rows:

Item Fat Content	Total_Sales
Low Fat	776319.68
Regular	425361.80

Insight: Low Fat products contribute significantly higher sales (₹776K) compared to Regular products (₹425K), indicating a clear customer preference towards healthier options.

C: Sales by Item Type

Query:

```sql
SELECT `Item Type`,
CAST(SUM(`Total Sales`)AS DECIMAL(10,2))AS Total_Sales
FROM blinkit_grocery_data
GROUP BY `Item Type`
ORDER BY Total_Sales DESC;
```

Output:

Result Grid | Filter Rows: | Expo

Item Type	Total_Sales
Fruits and Vegetables	178124.08
Snack Foods	175433.92
Household	135976.53
Frozen Foods	118558.88
Dairy	101276.46
Canned	90706.73
Baking Goods	81894.74
Health and Hygiene	68025.84
Meat	59449.86
Soft Drinks	58514.16
Breads	35379.12
Hard Drinks	29334.68
Others	22451.89
Starchy Foods	21880.03
Breakfast	15596.70
Seafood	9077.87

Insight: Fruits & Vegetables and Snack Foods lead sales, while categories like Seafood and Breakfast contribute the least, indicating a strong concentration of sales in a few key product types.

D: Fat Content by Outlet

Query:

```sql
SELECT `Outlet Location Type`,
    CAST(IFNULL(SUM(
                CASE
                WHEN `Item Fat Content` = 'Low Fat'
                THEN `Total Sales`
                END), 0) AS DECIMAL (10,2)) AS Low_Fat,
    CAST(IFNULL(SUM(
                CASE
                WHEN `Item Fat Content` = 'Regular'
                THEN `Total Sales`
                END), 0 ) AS DECIMAL (10,2)) AS Regular
FROM blinkit_grocery_data
GROUP BY `Outlet Location Type`
ORDER BY `Outlet Location Type`;
```

Output:

Result Grid | Filter Rows:

Outlet Location Type	Low_Fat	Regular
Tier 1	215047.91	121349.90
Tier 2	254464.77	138685.87
Tier 3	306806.99	165326.03

Insight: Sales increase from Tier 1 to Tier 3, with both Low Fat and Regular products performing strongest in Tier 3 outlets, indicating higher demand in larger or more developed locations.

E: Sales by Outlet Establishment Year

Query:

```sql
SELECT `Outlet Establishment Year`,
CAST(SUM(`Total Sales`)AS DECIMAL(10,2))AS Total_Sales
FROM blinkit_grocery_data
GROUP BY `Outlet Establishment Year`
ORDER BY `Outlet Establishment Year`;
```

Output:

Result Grid | Filter Rows:

Outlet Establishment Year	Total_Sales
1998	204522.26
2000	131809.02
2010	132113.37
2011	78131.56
2012	130476.86
2015	130942.78
2017	133103.91
2020	129103.96
2022	131477.77

Insight: Sales peak for older outlets (1998), while newer outlets show relatively consistent but lower performance, suggesting that established outlets tend to generate higher revenue over time.

F: Percentage of Sales by Outlet Size

Query:

```sql
SELECT `Outlet Size`,
CAST(
    SUM(`Total Sales`)
    AS DECIMAL(10,2))
    AS Total_Sales,
CAST(
    (SUM(`Total Sales`)* 100.0 /SUM(SUM(`Total Sales`))OVER())
    AS DECIMAL(10,2))
    AS Sales_Percentage
FROM blinkit_grocery_data
GROUP BY `Outlet Size`
ORDER BY Total_Sales DESC;
```

Output:

Result Grid | Filter Rows:

Outlet Size	Total_Sales	Sales_Percentage
High	248991.58	20.72
Medium	507895.73	42.27
Small	444794.17	37.01

Insight: Medium-sized outlets contribute the highest share of sales (42%), followed by small outlets (37%), while large outlets contribute the least, indicating that medium outlets are the most efficient revenue drivers.

F: Sales by Outlet Location

Query:

```
SELECT `Outlet Location Type`,
        CAST(
            SUM(`Total Sales`) AS DECIMAL(10,2)
            ) AS Total_Sales
FROM blinkit_grocery_data
GROUP BY `Outlet Location Type`
ORDER BY `Total_Sales` DESC;
```

Output:

Result Grid | Filter Rows:

Outlet Location Type	Total_Sales
Tier 1	336397.81
Tier 2	393150.64
Tier 3	472133.03

Insight: Tier 3 locations generate the highest sales, followed by Tier 2 and Tier 1, indicating stronger demand and revenue potential in higher-tier areas.

G: All Metrics by Outlet Type

Query:

```sql
SELECT `Outlet Type`,
CAST(SUM(`Total Sales`)AS DECIMAL(10,2))AS Total_Sales,
        CAST(AVG(`Total Sales`)AS DECIMAL(10,0))AS Avg_Sales,
        COUNT(*)AS No_Of_Items,
        CAST(AVG(`Rating`)AS DECIMAL(10,2))AS Avg_Rating,
        CAST(AVG(`Item Visibility`)AS DECIMAL(10,2))AS Item_Visibility
FROM blinkit_grocery_data
GROUP BY `Outlet Type`
ORDER BY Total_Sales DESC;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Outlet Type	Total_Sales	Avg_Sales	No_Of_Items	Avg_Rating	Item_Visibility
Grocery Store	151939.15	140	1083	3.98	0.10
Supermarket Type1	787549.89	141	5577	3.95	0.06
Supermarket Type2	131477.77	142	928	3.95	0.06
Supermarket Type3	130714.67	140	935	3.95	0.06

Insight: Supermarket Type1 dominates across all metrics with the highest sales and item count, making it the primary revenue driver, while other outlet types contribute significantly less despite having similar average sales and ratings.